ANSHUL PAL

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TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Python, Go, SQL (PostgreSQL), C, HTML5, CSS3
AI/ML & Data Science:	Algorithmic Recommender Systems, Collaborative Filtering, Markov Chains, Time-Decay Scoring, Feature Engineering, Heuristic Discovery
Full-Stack Frameworks:	React 18/19, Next.js (App Router), Node.js, Express, TipTap (ProseMirror), Tailwind CSS, Framer Motion, Web Audio API
Databases & Cloud Tools:	PostgreSQL, Supabase (RLS Auth), Prisma ORM, In-Memory TTL Caching, Git, Docker, RESTful APIs, Solana Web3

TECHNICAL PROJECTS

NeonWave – Algorithmic Music Recommendation & Streaming Platform github.com/anshul1729-in/Music-app
Node.js, Express, PostgreSQL, React, Web Audio API *Jun 2026*

- Architected a full-stack audio streaming platform powered by an algorithmic recommendation engine implementing personalized affinity scoring, discovery feeds, and track similarities.
- Engineered exponential time-decay scoring ($\lambda = \exp(-\Delta t / 7d)$) over user telemetry (play completion: +2.2, skip: −1.0, progress: +0.3), dynamically prioritizing fresh listener preferences.
- Formulated session-transition Markov chain modeling and item-item collaborative filtering using PostgreSQL Common Table Expressions (CTEs) and window functions (`LEAD`).
- Built an in-memory TTL cache (60,000ms) mitigating analytical database load, and developed an interactive 60fps audio visualizer via the Web Audio API.

Sanctuary – Full-Stack Knowledge Platform & Learning Analytics github.com/anshul1729-in/sanctuary
React 18, Supabase, PostgreSQL, TipTap ProseMirror, Tailwind CSS *Oct 2026*

- Engineered a full-stack daily knowledge platform with dual-storage architecture supporting synchronous local mode and cloud synchronization via Supabase PostgreSQL.
- Implemented robust multi-tenant data isolation and security using PostgreSQL Row Level Security (RLS) policies (`auth.uid() = user_id`) paired with JWT session caching.
- Integrated TipTap ProseMirror rich-text editor featuring 800ms debounced auto-save, dynamic domain taxonomy categorization, and a 16-week GitHub-style streak analytics engine.
- Engineered an object-relational mapping abstraction cleanly bridging camelCase client state with snake_case database records with optimistic UI updates.

Cendex – Non-Custodial Decentralized Exchange & Web3 Engine github.com/anshul1729-in/cendex
Next.js 14, TypeScript, Solana Web3, Prisma, PostgreSQL *Jul 2026*

- Built a non-custodial DEX platform integrating Jupiter aggregator API (`/v6/swap`) for real-time optimal routing across decentralized automated market makers (AMMs).
- Engineered client-side cryptographic transaction signing using `@solana/web3.js`, deserializing base64 payloads and executing Solana `VersionedTransaction` structures.
- Broadcasted raw transactions across high-throughput Helius RPC nodes with automated retry policies (`skipPreflight: true`) and validated blockhash/block height finality.
- Integrated NextAuth session security and Prisma ORM for relational PostgreSQL persistence of encrypted user wallet addresses and audit history.

Amigo-CLI – Moodle REST API Client & Study Automation github.com/anshul1729-in/amigo-cli
Python, REST APIs, Token Authentication, Automation *Sep 2026*

- Reverse-engineered Moodle Mobile REST API endpoints (`core_course_get_contents`, `mod_quiz_get_user_attempts`), bypassing sluggish browser interfaces.
- Built a modular CLI tool with persistent token caching in local configuration, enabling automated courseware downloading and real-time academic grade tracking.
- Automated quiz attempt aggregation and Markdown compilation, transforming raw API JSON review responses into structured offline study guides.

WORK EXPERIENCE

PropertyLay Jun 2025 – Sep 2025
Freelance Software Engineer *New Delhi, India*

- Architected and delivered a responsive commercial real estate web platform using Next.js (App Router), React, and Tailwind CSS for dynamic property portfolios.
- Engineered client-side routing, modular UI component hierarchies, and performant dynamic lead capture forms integrated with input validation pipelines.
- Optimized asset delivery, metadata, and Core Web Vitals, achieving sub-second Largest Contentful Paint (LCP) and zero-layout-shift UI transitions.

EDUCATION

Amity University Expected 2027

Bachelor of Computer Applications (BCA) – Computer Science *Noida, India*

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Web Development, Object-Oriented Software Engineering, Artificial Intelligence.